SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

SECURITY INCIDENT MAPPING (SIM) SYSTEM

SECTION 1: COVER PAGE & CHAPTER ONE

COVER PAGE

PROJECT TITLE

SECURITY INCIDENT MAPPING (SIM) SYSTEM

Course Code: SEN202

Submitted by:

Name: Emmanuel Samuel Inioluwa

Matric Number: 241201180

Department: Computer Science

Faculty: Computing

Institution: Adekunle Ajasin University

Submitted To:

Dr. O. O. Ajayi

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CHAPTER ONE

INTRODUCTION

1.1 Background of the Study
1.2 Security is one of the fundamental requirements for the growth and development of every educational institution. A secure learning environment promotes academic excellence, encourages innovation, and allows students and staff to carry out their daily activities without fear. However, insecurity has become a significant challenge in many higher institutions across Nigeria. Cases of theft, assault, burglary, harassment, phone snatching, vandalism, cult-related activities, and other criminal acts have continued to threaten the safety of students both on and off campus.

Many universities still rely on traditional methods of reporting security incidents. Victims are often expected to report incidents physically to security offices or school authorities. This process is usually slow, stressful, and discouraging, especially when incidents occur late at night or outside the campus premises. In some situations, victims may choose not to report crimes due to fear, inconvenience, or lack of confidence in the reporting process. Consequently, valuable information that could assist security agencies in preventing future occurrences is lost.

The rapid advancement of Information and Communication Technology (ICT) has created opportunities for institutions to improve security management through digital solutions. Web-based systems provide a convenient means for users to submit reports, receive updates, and communicate with administrators regardless of their location. Such systems improve accessibility, reduce reporting delays, and ensure that information is stored securely for future reference.

The Security Incident Mapping (SIM) System is a web-based application designed to improve the reporting and management of security incidents within the university community. The system enables students, staff, and landlords to report incidents electronically while allowing administrators to monitor reports, provide safety updates, manage emergency contacts, and coordinate security responses effectively.

By providing a centralized platform for reporting and managing incidents, the SIM system contributes to improving campus safety, increasing awareness, and promoting collaboration between members of the university community and security personnel.

1.3 Problem Statement

1.4 The increasing rate of insecurity within and around university campuses has become a major concern for students, staff, parents, and school authorities. Criminal activities such as theft, robbery, assault, harassment, and property damage often occur without being properly documented or reported. Many victims fail to report incidents because the existing reporting methods are inconvenient, time-consuming, or inaccessible.

Traditional reporting systems depend heavily on physical visits to security offices, which may not always be available during emergencies. Furthermore, paper-based records can easily become misplaced, damaged, or difficult to retrieve when needed. The absence of a centralized platform also makes it difficult for administrators to monitor trends, identify crime-prone areas, and provide timely safety information to the university community.

These challenges reduce the effectiveness of campus security management and delay responses to emergencies. Therefore, there is a need for a web-based solution that simplifies incident reporting, improves communication, enhances information management, and supports better decision-making regarding campus security.

1.3 Aim of the Project

The primary aim of this project is to develop a web-based Security Incident Mapping (SIM) System that provides an efficient platform for reporting, managing, and monitoring security incidents within the university community while improving communication between users and administrators.

1.5 Objectives of the Project
1.6 The objectives of the project are to:

Develop a secure web-based incident reporting platform.

Allow students, staff, and landlords to register and access the system.

Enable users to report security incidents quickly and conveniently.

Allow users to monitor the status of submitted reports.

Provide administrators with tools for managing reported incidents.

Enable administrators to publish safety announcements and emergency information.

Improve communication between users and campus security personnel.

Maintain a centralized database of reported security incidents.

Reduce delays associated with manual reporting procedures.

Improve overall campus security awareness.

1.5 Scope of the Project

The Security Incident Mapping (SIM) System focuses on providing an online platform for reporting and managing security incidents within the university environment. The system is intended for students, staff members, landlords, and authorized administrators.

The system covers user registration, login authentication, incident reporting, report monitoring, safety updates, emergency contacts, and administrative management of reports. Users can submit security complaints, view their reports, edit personal information, and receive important security announcements.

Administrators are responsible for reviewing reports, updating report status, managing users, publishing safety information, and maintaining emergency contact records.

The project is developed using HTML, CSS, and JavaScript as specified in the course requirements. Advanced technologies such as GPS tracking, artificial intelligence, SMS notifications, and live emergency response integration are outside the scope of this project but may be considered in future versions.

1.7 Benefits of the System

1.8 The implementation of the Security Incident Mapping System provides several benefits to both users and administrators.

Firstly, the system simplifies the process of reporting security incidents by allowing users to submit reports online without visiting the security office physically.

Secondly, it improves communication between students, landlords, staff members, and administrators by creating a centralized platform for information exchange.

The system also enables administrators to organize, monitor, and manage security reports more efficiently than traditional paper-based methods.

Another important benefit is the availability of safety announcements and emergency contacts, which helps users stay informed about security developments within the campus environment.

Furthermore, the system promotes accountability by maintaining digital records of submitted reports, making it easier to retrieve historical information whenever necessary.

The web-based nature of the application allows users to access the platform using different devices, making it convenient and accessible.

Finally, the SIM system contributes to creating a safer learning environment by encouraging timely reporting and improving the effectiveness of campus security management.

1.7 Definition of Terms

Security Incident: Any event that threatens the safety of individuals or property within or around the campus.

Incident Report: A formal record submitted by a user describing a security-related event.

Administrator: An authorized individual responsible for managing users, reports, and system content.

User: A registered student, staff member, or landlord who uses the system to report incidents and access security information.

Authentication: The process of verifying the identity of users before granting access to the system.

Database: A structured collection of information used to store reports and user records. In this project, browser local storage is used for data management.

Web-Based System: A software application that operates through an internet browser without requiring installation on the user's device.

Emergency Contact: Important contact information provided to users for quick access during emergency situations.

Safety Update: Official security announcements published by administrators to inform users about ongoing security matters.

Local Storage: A browser feature that allows data to be stored temporarily on the user's device for use within the application.

CHAPTER TWO

SOFTWARE REQUIREMENTS SPECIFICATION

The Software Requirements Specification (SRS) defines the functional and non-functional requirements of the Security Incident Mapping (SIM) System. It serves as a guide for the development of the system and outlines the services, constraints, and performance expectations of the application. The SIM system is designed as a browser-based application developed using HTML, CSS, and JavaScript to provide an efficient means of reporting and managing security incidents within the university environment.

2.1 Functional Requirements

Functional requirements describe the specific operations and services that the system must perform. These requirements explain how different users interact with the system to accomplish various tasks.

2.1.1 Authentication Module

The authentication module controls access to the system by ensuring that only registered users and administrators can access authorized features. Every user is required to create an account before accessing the reporting platform.

The authentication module shall provide the following functions:

User registration.

User login.

Administrator login.

Password verification.

Logout from the system.

Validation of user credentials.

During registration, users are required to provide personal information such as their full name, email address, username, phone number, and password. The system validates the information before creating an account. Existing users can log in using their registered username or email address together with their password.

Administrators are provided with a separate login page that grants access to management features unavailable to ordinary users.

2.1.2 User Registration

The system shall allow students, staff members, and landlords to create personal accounts.

During registration, the system shall:

Accept valid personal information.

Prevent duplicate usernames or email addresses.

Validate required fields.

Store user information securely.

Notify users upon successful registration.

Only registered users can access the reporting features of the application.

2.1.3 User Login

The login function enables registered users to access their dashboard.

The system shall:

Verify user credentials.

Deny access when invalid information is entered.

Display appropriate error messages.

Redirect successful users to their dashboard.

Allow users to log out securely after use.

2.1.4 Incident Reporting Module

The incident reporting module is the primary function of the SIM system. It allows users to notify administrators about security-related incidents occurring within or around the university community.

Users shall be able to:

Report theft.

Report assault.

Report robbery.

Report harassment.

Report vandalism.

Report suspicious activities.

Report missing property.

Report emergencies.

Each incident report shall contain:

Incident title.

Incident category.

Detailed description.

Date of occurrence.

Time of occurrence.

Location of incident.

Reporter’s identity.

Report submission date.

The system shall save every submitted report for future reference.

2.1.5 Report Management

After submitting a report, users should be able to monitor the progress of their complaints.

The system shall allow users to:

View submitted reports.

Check report status.

Read administrator feedback.

View report history.

Possible report status includes:

Pending

Under Investigation

Resolved

Closed

This functionality promotes transparency between users and administrators.

2.1.6 User Profile Management

The SIM system shall provide each user with a personal profile page.

Users shall be able to:

View profile information.

Edit profile details.

Update phone number.

Update email address.

Change password.

Save profile changes.

The system shall verify changes before updating stored information.

2.1.7 Safety Updates Module

The safety updates module enables administrators to communicate important security information to members of the university community.

Users shall be able to:

View safety announcements.

Read campus security tips.

Receive crime awareness information.

View public notices.

Access emergency alerts.

This feature helps improve awareness and encourages preventive measures against crime.

2.1.8 Emergency Contact Module

The system shall provide quick access to emergency contact information.

Information may include:

Campus Security Office.

University Health Centre.

Police Emergency Line.

Fire Service.

Ambulance Service.

Users can access these contacts whenever emergencies arise.

2.1.9 Administrator Functions

The administrator has complete control over the management of the SIM system.

The administrator shall be able to:

View all incident reports.

Search reports.

Filter reports by category.

Update report status.

Delete false or duplicate reports.

View registered users.

Delete inactive users.

Publish safety announcements.

Edit safety updates.

Manage emergency contacts.

Monitor system activities.

These functions enable administrators to effectively supervise the operation of the application.

2.2 Non-Functional Requirements

Non-functional requirements describe the quality standards that determine how efficiently the system performs its operations.

2.2.1 Performance Requirements

The system shall provide fast response times during user interactions. Pages should load within a reasonable period, and incident reports should be processed without unnecessary delays. The system should support multiple users accessing the application simultaneously while maintaining acceptable performance.

2.2.2 Reliability Requirements

The SIM system should operate consistently under normal conditions. Information entered by users should be stored correctly and remain available whenever needed. The system should minimize data loss and ensure continuous availability during operation.

2.2.3 Security Requirements

Security is essential because the system manages sensitive information.

The system shall:

Restrict administrator functions to authorized personnel.

Require authentication before access.

Protect user passwords.

Prevent unauthorized modification of reports.

Validate user input to reduce errors and malicious activities.

These measures help maintain confidentiality and integrity of information.

2.2.4 Usability Requirements

The interface should be simple, attractive, and easy to understand.

Users should be able to:

Navigate easily.

Locate important features quickly.

Submit reports without technical assistance.

Access information with minimal training.

The design should accommodate users with varying levels of computer literacy.

2.2.5 Maintainability Requirements

The system should be organized using clear HTML, CSS, and JavaScript structures to make future maintenance easier.

Developers should be able to:

Correct software errors.

Improve existing features.

Add new functionalities.

Update system components with minimal disruption.

2.2.6 Portability Requirements

The application should function properly on different operating systems and devices.

It should support modern browsers including:

Google Chrome

Microsoft Edge

Mozilla Firefox

Opera

The interface should also adapt to desktops, laptops, tablets, and smartphones.

2.2.7 Scalability Requirements

The system should accommodate future expansion without requiring complete redevelopment.

Future improvements may include:

GPS location tracking.

Email notifications.

SMS alerts.

Cloud database integration.

Mobile application support.

Artificial intelligence for crime prediction.

The design should allow these features to be incorporated when necessary.

2.3 Hardware Requirements

The minimum hardware required for the SIM system includes:

Desktop computer or laptop.

Smartphone or tablet.

Keyboard.

Mouse.

Stable internet connection.

Minimum 2 GB RAM.

At least 1 GB of available storage.

Display monitor with appropriate screen resolution.

These requirements are sufficient for both development and usage of the system.

2.4 Software Requirements

The SIM system requires the following software:

HTML5 for webpage structure.

CSS3 for interface styling.

JavaScript for application logic.

Visual Studio Code as the development environment.

Google Chrome or any modern web browser.

Browser Local Storage for temporary data storage.

Windows, Linux, or macOS operating system.

These software components provide the necessary tools for developing and operating the application efficiently.

SECTION 3

CONCLUSION

The Security Incident Mapping (SIM) System is a web-based application developed to improve the reporting and management of security incidents within the university community. The Software Requirements Specification (SRS) presented in this document clearly defines the functional and non-functional requirements necessary for the successful development and implementation of the system. These requirements serve as a blueprint that guides developers throughout the software development process while ensuring that the final product satisfies the needs of its intended users.

The proposed system provides a centralized platform through which students, staff members, landlords, and administrators can communicate security-related information efficiently. Instead of relying on slow and sometimes ineffective manual reporting methods, users are able to submit incident reports electronically through an intuitive web interface. This approach improves accessibility, minimizes reporting delays, and encourages members of the university community to report security concerns promptly.

The functional requirements identified in this specification demonstrate that the system supports essential operations such as user registration, secure authentication, incident reporting, profile management, report tracking, safety updates, emergency contact management, and administrative control. These functions work together to create a complete security reporting solution that promotes effective communication between users and campus security personnel.

Similarly, the non-functional requirements ensure that the SIM system performs efficiently while maintaining high standards of reliability, security, usability, portability, maintainability, and scalability. These quality attributes guarantee that the application remains dependable, easy to use, and adaptable to future technological improvements. Since the project is developed using HTML, CSS, and JavaScript, it provides a lightweight and browser-based solution that can be accessed on different devices without requiring complex installation procedures.

The implementation of this system offers numerous benefits to the university community. It improves record keeping through digital storage of incident reports, facilitates faster decision-making by administrators, enhances transparency in report management, and provides users with timely safety information and emergency contacts. These features collectively contribute to strengthening campus security and promoting a safer learning environment.

Furthermore, the modular design of the SIM system allows future enhancements without affecting its core functionality. Features such as geographical incident mapping, real-time notifications, cloud database integration, artificial intelligence for crime pattern analysis, and mobile application support can be incorporated as the system evolves to meet changing security demands.

In conclusion, the Security Incident Mapping (SIM) System represents a practical and effective application of software engineering principles to solving real-world security challenges within tertiary institutions. By combining modern web technologies with clearly defined software requirements, the system provides an efficient platform for reporting, monitoring, and managing security incidents. If properly implemented and maintained, it has the potential to improve campus safety, strengthen communication between stakeholders, encourage prompt reporting of incidents, and support the university's commitment to providing a secure environment for learning, teaching, and community living.

End of Software Requirements Specification (SRS)

Project Title: Security Incident Mapping (SIM) System

Student Name: Emmanuel Samuel Inioluwa

Matric Number: 241201180